AareML

Early Warning System for Swiss River Fisheries

Canton Zurich Innovation Sandbox — Phase III

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The Problem

"Every day during heat season we manually measure water temperature — and decide whether to launch a full fish rescue operation within 24 hours."

— Patrick Hager, President, Fischereiverein See + Gaster

1 Rescue decisions made manually, daily

Critical period: mid June – end of August

2 15-minute window to relocate fish

3 Known critical sites,
zero automated alerts

Today: daily manual checks → 24h rescue decision • Target: automated alert, 7 days in advance

7-day temperature early warning for Swiss river fisheries

Predict

AI model forecasts water temperature 7 days ahead at critical river locations

Alert

Mobile app notifies fisheries and AWEL when threshold is approaching

Act

Fish relocation within 15-minute window before conditions turn critical

Prototype: AareML — 14-day DO + temperature prediction validated on 86 Swiss gauges. Retraining for temperature focus and 7-day horizon.

We have the model. We need the data.

What we have

✓ Working AI model (AareML, validated on CAMELS-CH-Chem)

✓ Scientific partner — Eawag (NAWA FRACHT data)

✓ Domain partner — Fischereiverein See + Gaster (operational validation)

✓ Transfer learning: deploy at ungauged sites with only metadata

What the sandbox unlocks

→ Structured data access agreement with AWEL (150 cantonal monitoring sites)

→ Real-time AWEL temperature feeds (currently internal only)

→ Regulatory pathway: ML forecast → official fisheries advisory

→ PFAS monitoring integration (Phase 2)

Four complementary partners

Tatyana & Olga

AI & Machine Learning

AareML prototype, UBELIX HPC

Thiago Nascimento

Eawag

CAMELS-CH-Chem dataset co-author, NAWA FRACHT access, scientific validation

Patrick Hager

Fischereiverein See + Gaster

Domain expertise, go/no-go rescue ops

AWEL

Canton Zurich

Water quality monitoring authority, 150+ cantonal sites

Three specific asks

1 **Data MOU**

Structured access to AWEL real-time temperature feeds

2 **Regulatory pathway**

Framework for ML-based fisheries advisories

3 **PFAS pilot**

Early integration of emerging contaminant monitoring

Outcome: operational 7-day early warning system for Canton Zurich fisheries — first in Switzerland

github.com/polar-bear-after-lunch/AareML